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Smart Ultra Compact Inline Inspection for Sustainable Production

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Abstract

Pipelines are the veins of the Oil and Gas industry, a fixed asset with large capital costs and backbone for precious petroleum transportation. Precious as it is scarce and hard to find on the Earth's surface that's why it was called black gold years ago. Though, at first sight pipelines are less fascinating than the humongous offshore platforms and floating production systems. The fact with pipelines being the vena cava of the petroleum transfer body is that it consumes little energy for transportation as compared to oil tankers. Among the pipeline materials, carbon steel is the most preferable choice for contractors and owners as being economical, durable and environmentally friendly. However, it requires a bulk of operations for its smooth running to ensure the safe operation and manage its integrity [1].

This paper outlines the successful strategies adopted through the use of ultra compact in-line inspection technology to overcome long-standing integrity threats associated with the management of un-piggable flowline Assets. This successful technology trial marked the first in-line inspection of an oil flowline in ADNOC Onshore's operational history and represents a paradigm shift in how we manage the integrity of our oil flowline network in the future.

Background

ADNOC Onshore operates thousands unpiggable flowlines across various assets comprising different service types ranging sizes and has been experiencing severe internal corrosion in the oil flowlines, and one of the primary reasons for internal corrosion is the presence of stagnant corrosive fluid in the flowlines. Stagnant fluid settles at low points and results in accelerated localized corrosion. In addition, various external metal loss corrosion is also report due to lack of any mitigation to prevent the damage mechanism. All the Flowlines at various assets are designed either to ASME B 31.8/31.4 and installed as surface laid bare Carbon Steel above ground lines in normal terrain (sand dunes area) and at the sabkha area are laid over elevated supports (sleepers /hurdles) to clear off-the-ground contact.

ADNOC Onshore has been exploring various possibilities to overcome this problem to mitigate and improve the integrity issues developed during the operations. Various technologies were piloted and deployed to enhance the flowline integrity management. However, there was no any clear solution that fully enabled the or supported to draw a clear conclusion to arrive at the root cause and potential upgrades required in flowline design.

WHY Unpiggable?

Before identifying a technology solution or requirements, it is essential to first understand why a pipeline is considered unpiggable, although the term "unpiggable" is not legitimately familiar in dictionaries, it is commonly used within the pipeline industry to describe pipelines that cannot accommodate inline inspection (ILI) tools from start to finish. However, this term is increasingly viewed as misleading. In reality, any pipeline deemed "unpiggable" can potentially be made "piggable" through appropriate modifications to its design, access points, or operating conditions [2,3].

The key factors that dictate the feasibility of intelligent pigging operation are:

Pigging Facility: For any successful ILI, it requires an access point through which inspection tool can be launched and retrieved. Conventional methods rely on launchers and receivers, which may not exist on older or unmodified systems.

Design Considerations: The welded pipeline features, like any isolation valves, mitre bends, Offtakes or variations in bore passages can prevent passage of conventional ILI tools. Additionally, undocumented design changes or unknown modifications can introduce navigation obstacles for smooth inline inspection during the feasibility study.

Current Service Conditions: The Inspection tools largely rely on specific flow conditions to function properly when MFL technology is selected. If the flow is too low, too high, or if the internal media is too aggressive, it may affect the tool's performance or damage during operation.

In recent years, the industry has moved away from using the term "unpiggable" and instead, pipelines are now referred to as "difficult to inspect." The key difference is that these pipelines are not impossible to inspect, and they may just require more engineering effort, planning, or investment. For Instance:

- If a pipeline without pig traps, they can be installed.
- If a redesign is required or arrangements can be introduced post the feasibility study
- Operating conditions can be adjusted to suit the selected ILI tool requirements or through an external medium as prepollent (like Water, Diesel or Nitrogen).

The piggability of any challenging pipeline depends on both the feasibility and willingness to implement necessary modifications and mitigate operational challenges.

Key Challenges

Lack of Change documentation or Knowledge: Unlike conventional piggable pipelines designed for pigging, many challenging pipelines lacks as-built drawings, inspection records, or cleaning history. During the inspection planning without gauging or pre-inspection cleaning, unknown internal conditions, such as any navigation obstacles through bore restrictions or debris accumulations.

Pigging Facilities: Many of these pipelines are designed without any dedicated entry or exit points for ILI tools. A temporary set up or permanent pig traps or modified spool pieces may be required to enable the pigging.

Internal Surface Conditions: The internal surface cleanliness is critical for successful inspection when a UT based tool is equipped. The accumulated debris can affect with tool passage, tool velocity, and impact data quality of the inspection records.

Tool Selection: Selecting an appropriate inspection technology depends on both the inspection objective and pipeline conditions. Even if a tool can physically pass through the pipeline, poor internal conditions, such as heavy debris may result in unusable data.

The concept of an "unpiggable" pipeline is becoming obsolete. Every pipeline has the potential to be inspected, provided the right strategies are applied. By addressing challenges related to access, design, and operating conditions, pipelines previously considered difficult can be successfully inspected using modern tools and techniques.

Introduction to Technology

From an operator's standpoint, the risks tied to conventional inline inspection (ILI) tools are well-known and are manageable, this is not necessarily case for unconventional inspection solutions. Over the years, we have built confidence in their performance and understand the potential issues that can arise, even though there is genuine interest in adopting newer, more advanced solutions, there is often reluctance to be the first to take that step. One of the first questions we must address internally is, "What happens if the system fails?" If there is even a remote chance of the tool becoming stuck inside the pipeline, we need to see clear fail-safe mechanisms in place. Before committing, we typically expect the technology to undergo extensive trials and controlled testing to prove its capability and reliability. In addition, some practical challenges with the placement of ILI launchers and receivers, with some assets access is extremely congested with confined spaces adds complexity. These situations demand additional planning, stronger safety oversight, and careful execution [2].

At ADNOC Onshore, pigging valves were introduced for the first time in 2018 as part of our efforts to strengthen internal corrosion management. By modifying the flowline configuration to install valves at both the upstream and downstream ends, we were able to perform routine cleaning and de-watering pigging more efficiently. This step represented not just an operational improvement, but also a safer and more sustainable way of maintaining pipeline integrity (See Figure 1).

Figure 1—Pigging Valve Arrangement and Cleaning Pigging Tool

In 2024, a range of intelligent pipeline inspection vendors within the region and globally were reviewed and assessed for the small diameter inline inspection capabilities. Majority of flowlines are 6- inch in size, specifically looking into the inspection competencies focused around difficult- to-inspect pipelines where the internal condition of line is unknown were thoroughly reviewed for almost 15+ companies. Compact Inline Inspection technology trial utilizing Pigging Valves on above-mentioned flowline equipped with pigging valve was selected based on the pigging feasibility analysis conducted. The key feature in selecting the unique single bodied design is the tool's ability to navigate complex geometry and detect, locate and quantify metal loss anomalies.

Figure 2—Compact Inline Inspection Tool

Table 1—Flowline Candidate details selected for Technology Trial

Candidate Flowline Parameters	
Design code	*B 31.8*
Year of commissioning	*2013*
Pipeline Diameter	*6 NPS*
Pipeline Length	*4.74 km*
Pipeline Start Point	*Well Head*
Nominal Wall Thickness	*7.9 mm / 10.97 mm (road crossing)*
Product	*Well produced oil*
Design Pressure	*93 Bar g*
Design Temperature	*93° C*

Qualification Methodology

Like any technology trial, a clear set of systematic qualification criteria were established at each stage of the trial to ensure the acceptance of inline inspection results inline with international standards and bets practices comprising of the above-mentioned elements:

- Lost or Any Damage to Sensors After Run
- Sensor Noise or Data Loss
- Distance Inaccuracies
- Missed or Overlooked Features
- Velocity under-runs or Over-runs
- Location Benchmarking
- Signal data review and Results Accuracies Acceptance

Execution Strategy Adopted

Cleaning programs are not only important for supporting both the internal corrosion mitigation programs and preparation for in-line inspection. For non-conventional in-line inspections, line cleanliness can be one of the primary uncertainties, as pipeline debris can hinder the inspection's success and drastically increase its cost and complexity. Effective inspection requires adequate planning of cleaning programs, which typically includes review of the pipeline, assessment of the level of cleanliness required for the tool technology, and inline cleaning tool and chemical cleaning technology required for the line conditions. Preparation for a cleaning program should include a comprehensive review of prior cleaning runs, prior inspection runs, material product specifications considered when using cleaning products, and the potential effects of cleaning chemicals on the product stream.

Depending upon the type of inline cleaning tool utilized, many other attributes such as bore passage and bend requirements should also be considered to ensure that the inline inspection tool does not become stuck in the pipeline. Prior assessment shall be undertaken on how rigorous a cleaning program is required, determining if it should include several aggressive levels of inline cleaning tools [3].

The inspection activity was mainly phased into three major activities listed below using special set-up of water pumping units at the launcher side and waste disposal arrangements at receiver side of the flowline.

1. **Cleaning Pigging:** Mechanical cleaning to ensure cleanliness, and isolate the hydrocarbon from the pipeline
2. **Gauging Pigging:** Performed to ensure the tool passage throughout the pipeline and any bore reductions
3. **Inspection:** Actual run for inline inspection using the unique single body tool to detect any metal loss within the flowlines

Inspection Summary	
Retrieval & Data Recovery	*Successful*
Pig Receiving Condition	*No damage*
Run Time	*4 hrs 15 Minutes*
Inspection Coverage	*97%*
Full Length of Inspection	*5.25 Km*

Inspection Findings

The successful inspection of the flowline reported serval external corrosion anomalies, and internal corrosion anomalies across the pipeline length, with a maximum reported depth of 67% of Wall Thickness. The actual length of the flowline length reported from the inspection was 5.74 with max external and internal corrosion rate of 0.45 mm/year and 0.29 mm/year respectively[4].

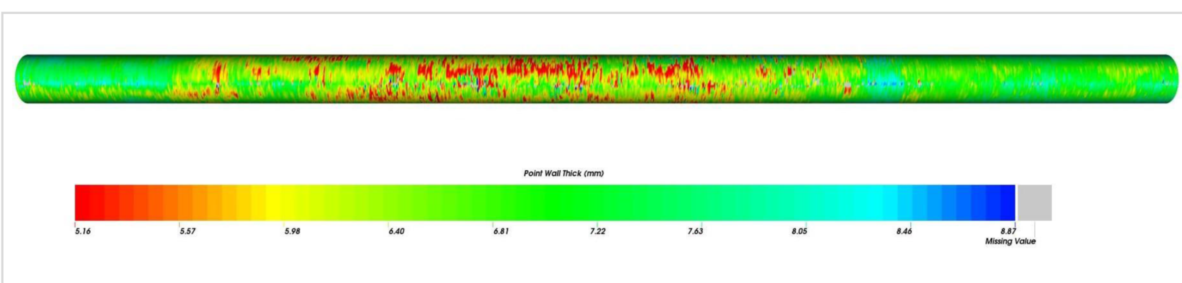

Figure 3—Inspection Finding at Severe Anomaly Location

Inspection Data Analysis

Acquisition of high-resolution inspection data through in-line inspection (ILI) technologies enables pipeline operators to transition from a predominantly reactive posture to a proactive integrity management paradigm. Such data provides the analytical foundation for optimizing inspection frequencies, conducting robust corrosion growth modeling, and formulating targeted repair and mitigation strategies. The precision of these insights ultimately drives substantial cost efficiencies, which can then be strategically reinvested to enhance the overarching integrity and operational resilience of the pipeline system[5].

Nevertheless, it is essential to situate inline inspection within the broader framework of a comprehensive integrity management program. An ILI campaign, though critical, constitutes only one element of a multifaceted initiative. Its value is maximized not merely through the resolution of the collected data, but also through the operational efficiency with which the inspection is executed and the ease of tool deployment and retrieval, reliability of passage, and minimal operational disruption. When integrated effectively, the resultant high-fidelity dataset underpins holistic fitness-for-service assessments and enables a more systematic, predictive, and sustainable approach to pipeline integrity management [4].

Although the flowline was expected to have significant internal metal corrosion mechanism, the results from inspection indicated that external metal loss corrosion was dominant corrosion mechanism, and below figures shown its significance of reported metal loss anomalies against the flowline profile and its orientation. In addition, further analysis on the significant external metal loss anomalies were reviewed to draw some conclusions. The aggressive soil corrosivity attributed to:

- Wetted soil by daily condensation or rainfall
- Contamination at certain locations due to integrity issues from adjacent flowlines
- Differential Aeration Corrosion Cells

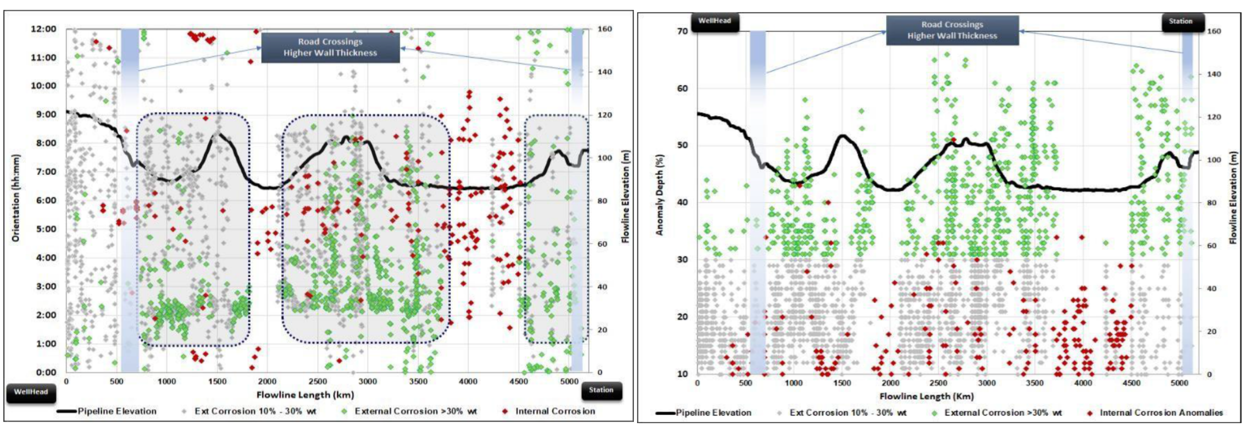

Figure 4—Orientation of Reported Metal Loss Anomalies and Significance of Reported Depth

The API 581 Estimated External Corrosion Rate is consistent with ILI measured corrosion rate at surface laid sections and no significant external corrosion anomalies reported at the Sabkha area (elevated sections) and road crossing locations.

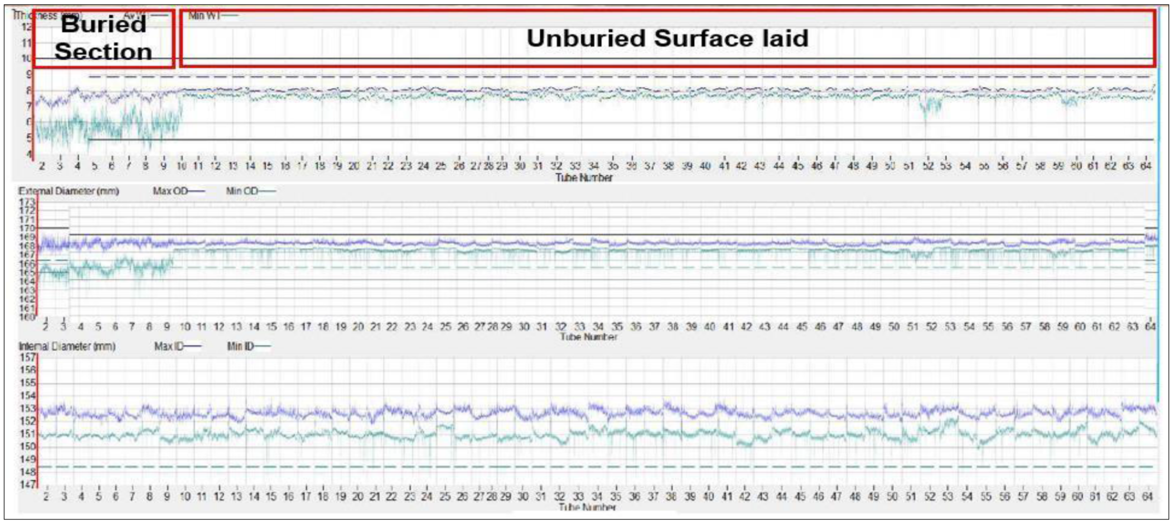

Figure 5—Classification of External Anomalies reported at Elevated Section

Results Verifications

Within the framework of in-line inspection (ILI) data validation, direct ultrasonic testing (UT) measurements are systematically employed as a reference benchmark against recorded ILI outputs. This comparative analysis serves a dual purpose: first, to substantiate the declared accuracy and performance envelope of the ILI technology, and second, to identify any deviations attributable to tool limitations, signal interpretation, or post-processing methodologies.

Discrepancies between UT and ILI datasets carry significant implications. Even minor inaccuracies may necessitate recalibration of the full ILI dataset, potentially prompting a comprehensive re-analysis of pipeline integrity. In more critical cases, such findings can catalyze further refinement of inspection technologies or analytical models to improve tool reliability and data precision. Consequently, the UT- to-ILI correlation process is not merely a quality assurance measure but a cornerstone of continuous improvement in integrity management practices, reinforcing both confidence in inspection results and the robustness of pipeline integrity assessments.

During the post–in-line inspection (ILI) data verification process, a targeted selection of anomalies – principally those corresponding to the deepest reported indications was identified for excavation and direct measurements. This selective validation strategy was undertaken to rigorously evaluate ILI tool performance and to ensure the robustness of the dataset employed for subsequent integrity assessments.

Excavation and field measurements confirmed strong concordance between the ILI-reported anomaly coordinates and the physical feature locations. Furthermore, dimensional evaluations demonstrated that all verified anomalies were within established sizing tolerances, thereby substantiating the accuracy claims of the inspection system. These findings not only validate the reliability of the specific ILI run but also provide empirical support for the broader application of high-resolution in-line inspection technologies as a credible foundation for integrity management and fitness-for-service evaluations.

Assessment Conclusions and Recommendations

Numerous recognized defect-acceptance and fitness-for-service (FFS) methodologies are currently employed for the evaluation of pipeline anomalies, including established standards and defect assessment inline with industrial practices. While these procedures provide rigorous calculation frameworks for assessing defect severity and establishing acceptance criteria, they alone are insufficient to deliver a comprehensive appraisal of pipeline integrity. The fidelity of such assessments is inherently constrained by uncertainties in the data input and by the broader engineering context in which the defects occur.

Critical factors influencing the reliability of an FFS evaluation include the tolerance and resolution of in-line inspection (ILI) datasets, the chronological age and material condition of the pipeline, and the physical properties of the transported product. Furthermore, operational and environmental loading conditions – such as cyclic pressure fluctuations, thermally induced stresses, and geotechnical influences including ground movement – must be systematically accounted for. Equally important is the precise characterization of the defect morphology and an understanding of its underlying causative mechanisms. Only through an integrated analytical approach that synthesizes these variables can a robust, defensible, and holistic assessment of pipeline condition and residual life be achieved. This successful inline inspection of un-piggable embarks a substantial shift in managing future flowline integrity for entire network. Utilizing this ultra compact in-line inspection technology will cater to inspect the majority of our flowline network and optimize future replacements.

The following key conclusions and recommendations are drawn from the smart inline inspection technology trail.

- Demonstrated tools' ability to navigate through various unknown sections and detection of anomalies in Oil flowline
- Technology is recommended for further use to enhance flowline integrity management by unlocking the smart replacements
- Technology can be used for corrosion growth monitoring to build actual corrosion models
- Inspection of representative sample flowlines is recommended to Integrate Ai Predictive Models
- Availability of inspection data for flowlines across various reservoir conditions will facilitate the ideal future design upgrade / material selection options

Challenges and Areas for Improvements

Despite the persistence of technical and operational challenges in in-line inspection inspections, recent advancements are progressively enabling the transition of previously classified as "unpiggable" pipelines to non-conventional inspection methodologies and ultimately toward compatibility with advanced inspection practices. Tool vendors have played a pivotal role in this evolution by developing and refining technologies that substantially mitigate, and in many cases eliminate, the extensive pipeline modifications historically required for inspection readiness [3].

With the increasing adoption of non-conventional inspection campaigns, novel technologies are being continually conceived, rigorously tested, and subsequently integrated into the body of industry best practice. Parallel innovations in cleaning protocols have yielded
significant improvements in line preparation, directly contributing to reduced inspection failure rates and enhanced operatio nal reliability. Ongoing research and technology development are directed toward a unifying objective: achieving a state in which both conventional and non-conventional ILI solutions can be deployed with comparable confidence, thereby strengthening the overall framework of pipeline integrity management and advancing industry standards of practice.

Overall technology trial experience provided significant positive outcomes, and identified some areas for future improvements that will enable real-time data analysis and integrity assessments [4]

- Tool to be equipped with accurate distance measuring features (like odometers)
- Pig tracking devices to enable precise pig location within the pipeline during inspection
- Integration INS (gyroscope/GIS) module will add significant value in post inspection data analysis/ integrity assessments

Value Proposition and Beyond the Trial

This innovative, advanced inspection solution is typically used for inspection of complex heater tubes in refineries, was dep loyed for oil flowlines in-line inspection for **the first time in the Middle East.** The tool inspection results were physically verified and was qualified for future utilization.

By employing this solution, ADNOC Onshore aims to achieve:

- Leak prevention, thereby minimizing the impact on HSE and Production.
- Sustainable design of oil flowline, based on reliable failure root cause analysis studies.
- Optimize future flowline replacement scope, leading to cost efficiency.

At ADNOC Onshore, we are committed to ensuring flowline integrity by applying cutting-edge technologies and leveraging AI solutions in support of our 2030 strategy.

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Disclaimer: All views/recommendations expressed in this Paper are of the Author(s) and may not necessarily be of the Author's employer

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